

Manan Mittal

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Summary

Machine Learning Engineer with hands-on experience in developing intelligent systems, started with generative AI applications on IBM Watsonx, engineering RAG pipelines integrated via APIs during a 3-month internship. Adept in applying deep learning and traditional machine learning methods to solve practical problems with clarity and precision. Strong analytical skills combined with a deep interest in building AI solutions that bridge the gap between complex algorithms and real-world impact. Looking forward on extending this foundation into reliable, production-grade AI deployment.

Education

B.Tech - VIT Bhopal University

CSE with specialization in Artificial Intelligence & Machine Learning

GPA: 9.31

Experience

AI Intern - IBM Generative AI

May 2024 - July 2024

- Built and deployed generative AI applications on the IBM Watsonx platform for real-world use cases.
- Implemented prompt engineering and RAG pipelines, integrating them via APIs into working solutions.
- Explored and documented LLM deployment strategies and integration patterns for client-facing applications.

Projects

IkigAI - Career Recommendation Engine (ML+LLM)

- Built and deployed a ML-based recommendation engine on a real-world dataset (200 samples, 15 features), comparing ensemble models to select the best-performing solution for end users.
- Achieved F1 scores: LightGBM (0.84), XGBoost (0.86), Random Forest (0.81).
- Owned the full workflow - feature engineering, encoding, model comparison, and evaluation - for production-ready output.

Pathwise - LLM + RL Career Advisory System

- Designed a real-time LLM-powered advisory system delivering context-aware career guidance through an interactive interface.
- Integrated reinforcement learning concepts to iteratively improve recommendation quality and personalization based on user feedback.
- Designed the system for adaptability across use cases, enabling data-driven decision support for early-stage students.

Image Captioning System (Deep Learning)

- Developed an end-to-end deep learning pipeline (ResNet50 + LSTM) that covers feature extraction, sequence modeling, and inference for automated image caption generation.
- Achieved BLEU-1 score of 0.514 and BLEU-4 score of 0.289 on benchmark datasets.
- Optimized preprocessing and inference pipeline for faster, deployment-ready performance.

Skills

Languages: Python, Java, C++

Engineering: REST API Integration, Model Deployment, Prompt Engineering, Data Pipelines, Git Version Control

ML/DL: Supervised Learning, CNN, RNN, LSTM, Recommendation Systems, GANs, LLMs, RAG

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV

Platforms/Tools: AWS, IBM Watsonx, Git, Google Colab, Hugging Face

Certifications

Oracle Cloud Infrastructure 2025 Certified Data Science Professional - cloud-based ML lifecycle management.

Applied Machine Learning(University of Michigan) - supervised-unsupervised learning and feature engineering.

IBM GenAI Watsonx - prompt engineering and RAG pipeline design.

Java (HackerRank Gold Badge) - object-oriented programming and data structures proficiency.

NPTEL Cloud Computing - cloud architecture and virtualization fundamentals.

NPTEL Introduction to IoT - IoT architecture and sensor-network fundamentals.

Achievements

- OSC AI Build 1.0 - Achieved 6th rank among 3000+ participants *Demo*
- Fuel Blend Prediction,NASSCOM Shell AI Hackathon(Top 10%): Processed 100K+ records, built ensemble ML models, reduced runtime to under 5 minutes.
- 2x State & National Runner-Up, 1x State Champion -UCMAS Mental Arithmetic Competition.
- Qualified GATE (CS) 2026 with a GATE score of 443.